Key Terms Used in TF-IDF

Before diving into the calculations and code, it's essential to understand the key terms:

- **t**: term (word)
- **d**: document (set of words)
- **N**: count of corpus
- **corpus**: the total document set

What is Term Frequency (TF)?

The frequency with which a term occurs in a document is measured by term frequency (TF). A term's weight in a document is directly correlated with its frequency of occurrence. The TF formula is:

$$\mathrm{TF}(t, d) = \frac{\text{count of } t \text{ in } d}{\text{number of words in } d}$$

What is Document Frequency (DF)?

The significance of a document within a corpus is gauged by its Document Frequency (DF). DF counts the number of papers that contain the phrase at least once, as opposed to TF, which counts the instances of a term in a document. The DF formula is:

DF(t)=occurrence of t in documents

What is Inverse Document Frequency (IDF)?

The informativeness of a word is measured by its inverse document frequency, or IDF. All terms are given identical weight while calculating TF, although IDF helps scale up uncommon terms and weigh down common ones (like stop words). The IDF formula is:

$$\mathrm{IDF}(t) = \log\left(\frac{N}{\mathrm{DF}(t)+1}\right)$$

where N is the total number of documents and DF(t) is the number of documents containing the term t.

What is TF-IDF?

TF-IDF stands for Term Frequency-Inverse Document Frequency, a statistical measure used to evaluate how important a word is to a document in a collection or corpus. It combines the importance of a term in a document (TF) with the term's rarity across the corpus (IDF). The formula is:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \log\left(\frac{N}{\text{DF}(t)+1}\right)$$

Numerical Calculation of TF-IDF

Let's break down the numerical calculation of TF-IDF for the given documents:

Documents:

1. "The sky is blue."
2. "The sun is bright today."
3. "The sun in the sky is bright."
4. "We can see the shining sun, the bright sun."

Step 1: Calculate Term Frequency (TF)

Document 1: "The sky is blue."

Term	Count	TF
the	1	1/4
sky	1	1/4
is	1	1/4
blue	1	1/4

Document 2: "The sun is bright today."

Term	Count	TF
the	1	1/5
sun	1	1/5
is	1	1/5
bright	1	1/5
today	1	1/5

Document 3: "The sun in the sky is bright."

Term	Count	TF
the	2	2/7
sun	1	1/7
in	1	1/7
sky	1	1/7
is	1	1/7
bright	1	1/7

Document 4: “We can see the shining sun, the bright sun.”

Term	Count	TF
we	1	1/9
can	1	1/9
see	1	1/9
the	2	2/9
shining	1	1/9
sun	2	2/9
bright	1	1/9

Step 2: Calculate Inverse Document Frequency (IDF)

Using N=4N = 4N=4:

Term	DF	IDF
the	4	log(4/4+1)=log(0.8)≈−0.223
sky	2	log(4/2+1)=log(1.333)≈0.287
is	3	log(4/3+1)=log(1)=0
blue	1	log(4/1+1)=log(2)≈0.693
sun	3	log(4/3+1)=log(1)=0
bright	3	log(4/3+1)=log(1)=0
today	1	log(4/1+1)=log(2)≈0.693
in	1	log(4/1+1)=log(2)≈0.693
we	1	log(4/1+1)=log(2)≈0.693
can	1	log(4/1+1)=log(2)≈0.693
see	1	log(4/1+1)=log(2)≈0.693
shining	1	log(4/1+1)=log(2)≈0.693

Step 3: Calculate TF-IDF

Now, let’s calculate the TF-IDF values for each term in each document.

Document 1: “The sky is blue.”

Term	TF	IDF	TF-IDF
the	0.25	-0.223	0.25 * -0.223 ≈-0.056
sky	0.25	0.287	0.25 * 0.287 ≈ 0.072
is	0.25	0	0.25 * 0 = 0
blue	0.25	0.693	0.25 * 0.693 ≈ 0.173

Document 2: “The sun is bright today.”

Term	TF	IDF	TF-IDF
the	0.2	-0.223	0.2 * -0.223 ≈ -0.045
sun	0.2	0	0.2 * 0 = 0
is	0.2	0	0.2 * 0 = 0
bright	0.2	0	0.2 * 0 = 0

today	0.2	0.693	0.2 * 0.693 ≈0.139

Document 3: "The sun in the sky is bright."

Term	TF	IDF	TF-IDF
the	0.285	-0.223	0.285 * -0.223 ≈ -0.064
sun	0.142	0	0.142 * 0 = 0
in	0.142	0.693	0.142 * 0.693 ≈0.098
sky	0.142	0.287	0.142 * 0.287≈0.041
is	0.142	0	0.142 * 0 = 0
bright	0.142	0	0.142 * 0 = 0

Document 4: "We can see the shining sun, the bright sun."

Term	TF	IDF	TF-IDF
we	0.111	0.693	0.111 * 0.693 ≈0.077
can	0.111	0.693	0.111 * 0.693 ≈0.077
see	0.111	0.693	0.111 * 0.693≈0.077
the	0.222	-0.223	0.222 * -0.223≈-0.049
shining	0.111	0.693	0.111 * 0.693 ≈0.077
sun	0.222	0	0.222 * 0 = 0
bright	0.111	0	0.111 * 0 = 0